

T3 BSM UV terms grouped by field content

EFT order: 5; loop order: 1. Columns are determined only from BSM UV terms. Empty cells denote no UV term with that field combination.

Field combinations 1 of 3

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	F	S_1	S_2	$F\ D$
T3-d3-d5-F4	0	3	0	5	1	4	$\frac{1}{2}$	$-\overline{N_{j_1,r_1}}\,N_{j_1,r_1}\,M_F$	$\begin{aligned} & -N_0\,\text{wfr}\,\text{color}\,1_{12}\,\text{NewStr}\,\text{color}\,1_{12}^{\dagger}\,H_F^0 \\ & +\frac{1}{2}\text{amb}\,\text{ds}\,\text{Hfr}\,1N_0\,\text{wfr}\,\text{color}\,1_{12}\,\text{NewStr}\,\text{color}\,1_{12}\,\text{Tr}\,1\,\text{Str}\,\text{Hfr}\,1N_0\,\text{GfHfr}\,1_{12},120,120\,\text{NewStr}\,\text{color}\,1_{12}^{\dagger}\,\text{NewStr}\,\text{color}\,1_{12} \\ & +\frac{1}{2}\text{amb}\,\text{ds}\,\text{Hfr}\,1N_0\,\text{wfr}\,\text{color}\,1_{12}\,\text{NewStr}\,\text{color}\,1_{12}\,\text{Tr}\,1\,\text{Str}\,\text{Hfr}\,1N_0\,\text{GfHfr}\,1_{12},120,120\,\text{NewStr}\,\text{color}\,1_{12}^{\dagger}\,\text{NewStr}\,\text{color}\,1_{12} \end{aligned}$	$\begin{aligned} & -H_F^0\,N_0\,\text{wfr}\,\text{color}\,1_{12}\,N_0\,\text{wfr}\,\text{color}\,1_{12}^{\dagger} \\ & +\frac{1}{2}\text{amb}\,\text{ds}\,\text{Hfr}\,1N_0\,\text{wfr}\,\text{color}\,1_{12}\,\text{NewStr}\,\text{color}\,1_{12}\,\text{Tr}\,1\,\text{Str}\,\text{Hfr}\,1N_0\,\text{GfHfr}\,1_{12},120,120\,\text{NewStr}\,\text{color}\,1_{12}^{\dagger}\,\text{NewStr}\,\text{color}\,1_{12} \\ & +\frac{1}{2}\text{amb}\,\text{ds}\,\text{Hfr}\,1N_0\,\text{wfr}\,\text{color}\,1_{12}\,\text{NewStr}\,\text{color}\,1_{12}\,\text{Tr}\,1\,\text{Str}\,\text{Hfr}\,1N_0\,\text{GfHfr}\,1_{12},120,120\,\text{NewStr}\,\text{color}\,1_{12}^{\dagger}\,\text{NewStr}\,\text{color}\,1_{12} \end{aligned}$	$+i\overline{N_{j_1,r_1}}\,\gamma^{\mu 1}\,D_{\mu 1}\,(N_{j_1,r_1})$

Field combinations 2 of 3

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	S_1	D	$S_1\ H$	$S_1\ S_2$	$S_2\ D$
T3-d3-d5-F4	0	3	0	5	1	4	$\frac{1}{2}$	$+D_{\mu_1}$	$(\text{NewScalar1}_{j_1})^\dagger D_{\mu_1} (\text{NewScalar1}_{j_1})$	$\begin{aligned} &+\frac{1}{2}\text{Im bdaH1Inv}[\text{NewScalar1}_{j_2}\text{T3HiggsPortal1CGInv1}_{\alpha_1,\alpha_2,j_1,j_2}H_{\alpha_1}]^\dagger\text{NewScalar1}_{j_1}^\dagger \\ &+\frac{1}{2}\text{Im bdaH1Inv}[\text{NewScalar1}_{j_2}\text{T3HiggsPortal1CGInv1}_{\alpha_1,\alpha_2,j_2,j_1}H_{\alpha_1}]^\dagger\text{NewScalar1}_{j_1}^\dagger \\ &+\frac{1}{2}\text{Im bdaH1Inv}[\text{NewScalar1}_{j_2}\text{T3HiggsPortal1CGInv2}_{\alpha_1,\alpha_2,j_1,j_2}H_{\alpha_1}]^\dagger\text{NewScalar1}_{j_1}^\dagger \\ &+\frac{1}{2}\text{Im bdaH1Inv}[\text{NewScalar1}_{j_2}\text{T3HiggsPortal1CGInv2}_{\alpha_1,\alpha_2,j_2,j_1}H_{\alpha_1}]^\dagger\text{NewScalar1}_{j_1}^\dagger \end{aligned}$	$+ \text{Im bdaH1Inv}[\text{NewScalar1}_{j_2}\text{NewScalar1}_{j_2}\text{GrassmannCGInv1}_{i_1,i_2,\alpha_1,\alpha_2}\text{NewScalar1}_{i_1}]^\dagger\text{NewScalar1}_{i_1}^\dagger$	$+D_{\mu_1} (\text{NewScalar2}_{j_1})^\dagger D_{\mu_1} (\text{NewScalar2}_{j_1})$

Field combinations 3 of 3

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$S_2 H$	$S_1 F \ell$	$S_1 S_2 H$	$S_2 F \ell$
T3-d3-d5-F4	0	3	0	5	1	4	$\frac{1}{2}$	$+i\text{am bdaH2Inv1}H_{a_2}\text{NewScalar}2_{j_2}\text{T3Hgg1Partial2CGI}m^{i_{a_1},a_2,j_1,j_2}H_{a_1}{}^{\dagger}\text{NewScalar}2_{j_1}{}^{\dagger}$	$+ \overline{\ell_{a_1,i_1}} C P_R \overline{N_{j_1,r_1}}^T \text{NewScalar}1_{j_1}\text{T3Y1CG}_{a_1,j_1,j_1}y^{1,i_1,r_1}$	$+ \overline{\lambda_{\tau 2}} \text{NewScalar}2_{j_1}\text{T3MixCG}_{a_1,a_2,j_1,j_1}H_{a_1}{}^{\dagger}H_{a_2}{}^{\dagger}\text{NewScalar}1_{j_1}{}^{\dagger}$	$+ \overline{y^{2,i_1,r_1}} \overline{N_{j_1,r_1}} P_L \ell_{a_1,i_1} \text{NewScalar}2_{j_1}\text{T3Y2CG}_{a_1,j_1,j_1}$
								$+i\text{am bdaH2Inv2}H_{a_2}\text{NewScalar}2_{j_2}\text{T3Hgg1Partial2CGI}m^{2_{a_1},a_2,j_1,j_2}H_{a_1}{}^{\dagger}\text{NewScalar}2_{j_1}{}^{\dagger}$	$+ \overline{y^{1,i_1,r_1}} \overline{\ell_{a_1,i_1}}^T C P_L N_{j_1,r_1} \text{T3Y1CG}_{a_1,j_1,j_1} \text{NewScalar}1_{j_1}{}^{\dagger}$	$+ H_{a_1} H_{a_2} \text{NewScalar}1_{j_1}\text{T3MixCG}_{a_1,a_2,j_1,j_1} \lambda_{\tau 2} \text{NewScalar}2_{j_1}{}^{\dagger}$	$+ \overline{\ell_{a_1,i_1}} P_R N_{j_1,r_1} \text{T3Y2CG}_{a_1,j_1,j_1}y^{2,i_1,r_1} \text{NewScalar}2_{j_1}{}^{\dagger}$